



## SIMATS ENGINEERING

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**Course Code:CSA0305**

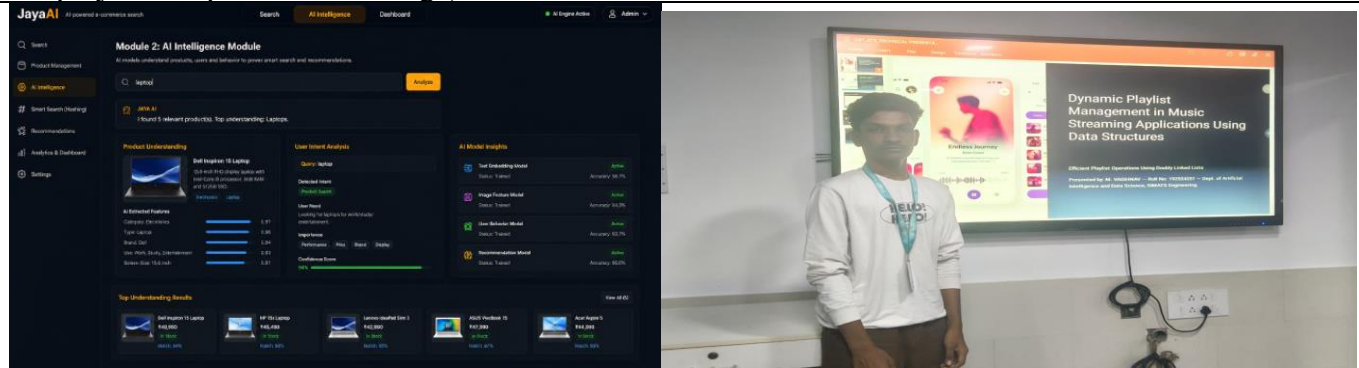
**Slot: A**

**Course Name:Data Structure**

**Course Faculty: Dr Uma Priyadarsini**

**Project Title: Implementation of an AI-Powered Penetration Testing Assistant for Intelligent Vulnerability Assessment and Security Recommendations.**

**Module Photographs:** (3 photographs –Module Photo, Individual student contribution module work in the project and presentation image)



**Project Description: Implementation of an AI-Powered Penetration Testing Assistant for Intelligent Vulnerability Assessment and Security Recommendations.**

The **AI-Powered Penetration Testing Assistant** is implemented to automate and improve the process of vulnerability assessment and security analysis. The system combines **Artificial Intelligence (AI)** with commonly used penetration-testing tools to collect, analyze, and interpret security information from an authorized target environment. Initially, reconnaissance information such as IP addresses, open ports, running services, server details, and web technologies is collected using tools such as **Nmap, WHOIS, WhatWeb, Ping, and Nikto**. The collected scan results are then passed to the AI module for intelligent analysis.

The AI analyzes the obtained results and identifies potential vulnerabilities, misconfigurations, outdated services, exposed directories, and insecure security settings. Each vulnerability is classified according to its **severity level**, such as Low, Medium, High, or Critical.

The assistant can be developed using **Python**, which is used to automate tool execution, process scan outputs, and communicate with the AI model. A simple interface can be provided to display the scan results, vulnerability details, severity levels, and recommended solutions. Finally, the system generates a structured security report containing the **target information, identified vulnerabilities, risk levels, observations, and remediation recommendations**.

**Student Signature**

**Guide Signature**

